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Group 35

Tasks & Contributions

I have mainly contributed to the code. To understand the logic of the DBMS, I tried to learn from the lectures and the internet. After writing the code, I have added several tests and tries many edge cases which I have found so far. In this way, I found many unwanted results, and the longest part was the debugging, especially I/O counts issues.

Also, I added a mini script for the experiments in the record.txt. So, we handled those experiments smoothly thanks to the ready commands. In addition, I had responsibility to create and check Git repository.

Collaboration & Challenges

At first, we discussed the strategies and options to choose together. Then, we implemented those selections into the code. The hardest part for me was I/O counts debugging. I had several issues for that. Also, Bplus-tree node splitting was little bit challenging, distinguishing a leaf split (separator key copied up) from an internal split (key pushed up and removed from the node). We resolved it by working the cases out by hand and rewriting the split functions until they were correct.

Self-Assessment

I have learned the fundamentals of the DBMS engine, and how our queries work. I am now familiar the database logic. Also, I have learned the strategies, Bplus_tree, hash_index and heap_scan. The assignment was actually quite hard, but it teaches me so much.

Use of AI Tools

I have used Claude for code and Gemini for learning. To be honest, since the project is quite comprehensive and hard, I asked many questions to Gemini and write the code by the help of Claude. However, I have never copied pasted the code blocks. I always looked and understood the code firstly, then implemented them. Also, I used Claude for debugging. Especially, I had hard time to debug the I/O counts issues, so I asked Claude where the problem is. Since the issue was big, Claude also had hard time to determine the bug and we brainstormed together :) Also, to understand better, I wanted Claude to write docstrings and comments to the code and keep them up to date.